

Test Strategy:

Scope: include patient appointment booking

1. View available slots within 30 days.
2. Create booking.
3. Prevent booking conflict.
4. Cancel booking a per 24h policy.
5. Patient can have 3 upcoming booking max.
6. Email and SMS when booking and cancel.
7. Cross platforms test (Android, IOS, and Angular).
8. Time zone support.

Risk Areas:

1. Concurrent Booking: 2 users booking same slot in the same time.
2. Time zone: patient in time zone different than provider time zone.
3. Permission: patient can show only his appointment.
4. Upcoming appointments limit: patient can have 3 upcoming appointments max.
5. Notifications: email/SMS not sent.
6. Cancellation roles: maybe some roles are fail.

Testing types:

1. Functional test: to validate all acceptance criteria and make sure system roles are working well such (prevent more than 3 upcoming appointment, cancellation within 24h)
2. Boundary value testing: this feature contain a lot of boundaries and we should test exact boundary value and less than 1 and more than 1.
3. Negative Testing: like book unavailable appointment, cancel too late, and book in past.
4. Authorization Testing: to make sure each user show only his data.
5. Integration Testing: to test SMS/Email if sent or not.
6. Cross-Platform Testing to make sure mobile and web have same behavior and same goal.
7. API automation for all booking and cancellation operations to make regression easier when we re test defect.

Testing data and HIPAA consideration:

We will create our own dummy data and will use QA server, never use any production data. We will create dummy patient accounts in several cases (reach 3 booking limit, have a booking within 24h, time zone accounts) also will create dummy provider accounts

Test cases:

ID	Title	Type	Priority	Preconditions	Steps	Expected result
TC-1	View available slots within 30 days	Functional	high	Patient logged in	Choose provider	Available slots for next 30 days displayed
TC-2	Book available slot successfully	Functional	critical	Patient logged in	Select slot > book	Appointment created. Slot not available any more. Confirmation shown
TC-3	Prevent double booking	Functional	critical	Patient A and B logged in	Patient A selects X slot and Patient B select same slot. A press book then B press book	Patient A: Appointment created. Slot not available any more. Confirmation shown Patient B: Error message will be shown "booking not available any more"
TC-4	Cannot book past appointment	Negative	high	Past slot exist	Choose slot in past and press book	User can not choose past slot OR when press book error message will shown "can not choose past slot"
TC-5	Book appointment	Boundary	high	Slot at day 30 exists	Book slot	Booking succeeds

	nt exactly 30 days ahead					
TC-6	Cannot book beyond 30 days	Boundary	high	Slot at day 31 exists	Book slot	Booking rejected
TC-7	Cancel appointment more than 24h before	Functional	high	Appointment after 25h	Cancel appointment	Cancellation succeeds and notifications sent
TC-8	Cannot cancel within 24h	Boundary	high	Appointment after 23h	Cancel appointment	Cancellation failed and error message shown
TC-9	Maximum 3 upcoming appointments allowed	Boundary	critical	Patient has 3 future appointments	Book fourth appointment	Booking rejected
TC-10	Booking notification sent	Integration	high	Valid booking	Create booking	Email and SMS delivered
TC-11	Timezone conversion patient vs provider	TimeZone	critical	Patient TZ GMT+3 Provider TZ GMT +1	Create booking 9 PM	Patient show the booking time 9 PM, Provider show the booking time at 7PM
TC-12	Consistent booking flow on Web, Android, and iOS	Cross-Platform	high	Accounts available	Perform booking on all platforms	Same behavior and results across platforms
TC-13	Patient cannot access another patient's	Security	critical	Accounts available	Patient A logged in to system	Patient B should not show anything related to

	appointme nt				and logged out then Patient B logged in to system using same device	Patient A
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